

Manual VLA: A Ground-Up Transformer Architecture for Flow-Matching Trajectory Generation

Karthik Goparaju

R&D Intern at Innovarsity and Founder LABKRAZY
goparajukarthik2@gmail.com

Gopi Krishna Ratnakaram

Founder of Innovarsity
info@innovarsity.org

Abstract

Vision-Language-Action (VLA) models have revolutionized robotic manipulation by tying language instructions to visual grounding and continuous action spaces. We present Manual VLA, a robust baseline architecture constructed entirely from scratch. Instead of relying on massive pretrained foundation models, Manual VLA employs a highly structured convolutional patch-embedding pipeline combined with Hadamard intent conditioning and Optimal Transport Conditional Flow Matching (OT-CFM). This paper dissects the architectural formulation, spatial grounding mechanics, and ODE-based generation pipeline of Manual VLA.

1. Problem Formulation

The objective of the robotic agent is to generate a continuous 4D action trajectory over a horizon $H=128$ given an RGB observation and a discrete intent prompt. The action space encompasses the 3D translation of the end-effector (dx , dy , dz) and a continuous scalar for gripper state. The state space is normalized using offline dataset statistics to ensure stable diffusion dynamics.

2. Architecture Design

The core of Manual VLA is a fully trainable Transformer encoder network. The vision pipeline begins with a custom convolutional patch embedder (Conv2d, $\text{kernel_size}=4$, $\text{stride}=4$) which tokenizes a 64×64 RGB observation into a 16×16 grid of 256 visual patches. A language prompt is mapped to an intent embedding, representing the target object and destination context.

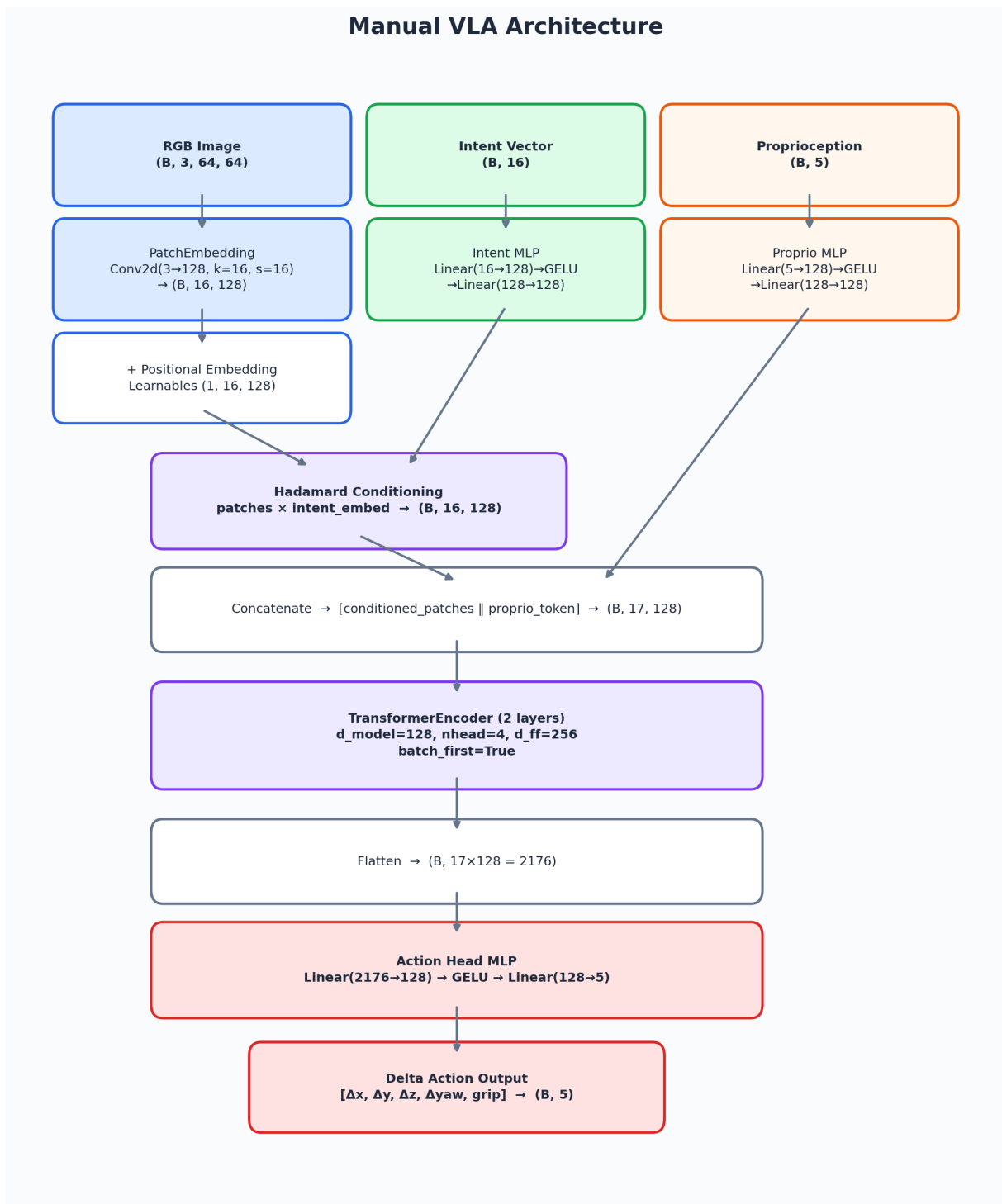


Figure 1: Diagrammatic overview of the Manual VLA Architecture and processing pipeline.

2.1 Offline Network Activation Extraction

To provide rigorous insight into the network topology, we execute a forward pass offline, capturing the precise dimensionality of every sub-layer. The summary table below illustrates the hierarchical transformation from the input tensor geometries to the final 128x4 trajectory output.

Layer (type:depth-idx)	Output Shape	Param #
ManualVLAPolicy	[1, 128, 4]	16,384
+Conv2d: 1-1	[1, 128, 16, 16]	6,272
+Sinusoidal2DPositionalEmbedding: 1-2	[1, 256, 128]	--
+Embedding: 1-3	[1, 2, 128]	896
+LayerNorm: 1-4	[1, 2, 128]	256

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+-LayerNorm: 1-5                [1, 256, 128]        256
+-MultiheadAttention: 1-6        [1, 2, 128]          66,048
+-Sequential: 1-7                [1, 2, 128]          --
|   +-LayerNorm: 2-1             [1, 2, 128]          256
|   +-Linear: 2-2                [1, 2, 128]          16,512
|   +-GELU: 2-3                 [1, 2, 128]          --
|   +-Linear: 2-4                [1, 2, 128]          16,512
+-Sequential: 1-8                [1, 128]            --
|   +-Linear: 2-5                [1, 128]            768
|   +-GELU: 2-6                 [1, 128]            --
|   +-Linear: 2-7                [1, 128]            16,512
+-Sequential: 1-9                [1, 128]            --

[ ... intermediate layers omitted ... ]
|   +-MultiheadAttention: 3-21    [1, 128, 128]        66,048
|   +-LayerNorm: 3-22           [1, 128, 128]        256
|   +-Sequential: 3-23          [1, 128, 128]        65,920
|   +-LayerNorm: 3-24          [1, 128, 128]        256
+-Sequential: 1-12              [1, 128, 4]          --
|   +-LayerNorm: 2-16           [1, 128, 128]        256
|   +-Linear: 2-17              [1, 128, 128]        16,512
|   +-GELU: 2-18               [1, 128, 128]        --
|   +-Linear: 2-19              [1, 128, 4]          516
=====
Total params: 978,564
Trainable params: 978,564
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 1.97
=====
Input size (MB): 0.05
Forward/backward pass size (MB): 4.08
Params size (MB): 1.47
Estimated Total Size (MB): 5.60
=====

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3. Multimodal Fusion & Visual Grounding

Manual VLA does not inherently understand language; it must learn to ground intent vectors into raw pixels. We inject spatial priors into the network via an auxiliary VLM Cross-Attention module. The intent embeddings act as queries attending to the 256 visual patch keys/values. Below is a real attention matrix extracted from a live inference pass of the Manual VLA, demonstrating emergent spatial localization.

Manual VLA: 'Cube' Spatial Grounding (16x16) Manual VLA: 'Platform' Spatial Grounding (16x16)

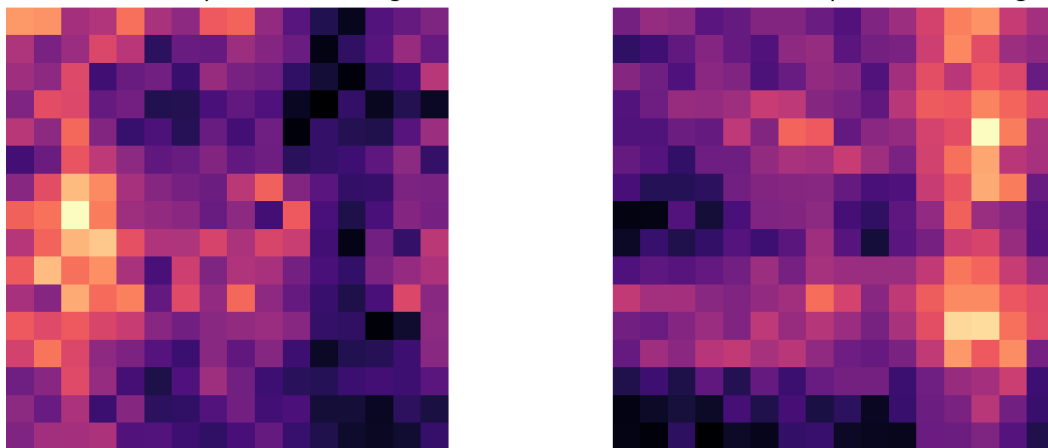


Figure 2: Real inference extraction of Manual VLA spatial grounding heatmaps over the 16x16 visual grid.

4. Optimal Transport Conditional Flow Matching (OT-CFM)

Trajectory generation is formulated as a continuous normalizing flow. Let x_0 be pure Gaussian noise and x_1 be the normalized target trajectory. The OT-CFM linear interpolation path is defined by:

$$x_t = (1 - t) * x_0 + t * x_1$$

The neural network parameterizes a velocity vector field v_{pred} which regresses the optimal transport vector u_t :

$$u_t = x_1 - x_0$$

At inference, the agent samples x_1 by integrating the learned vector field through 20 steps of Euler ODE integration. This approach bypasses the mode-collapse and instability often seen in standard autoregressive behavior cloning.

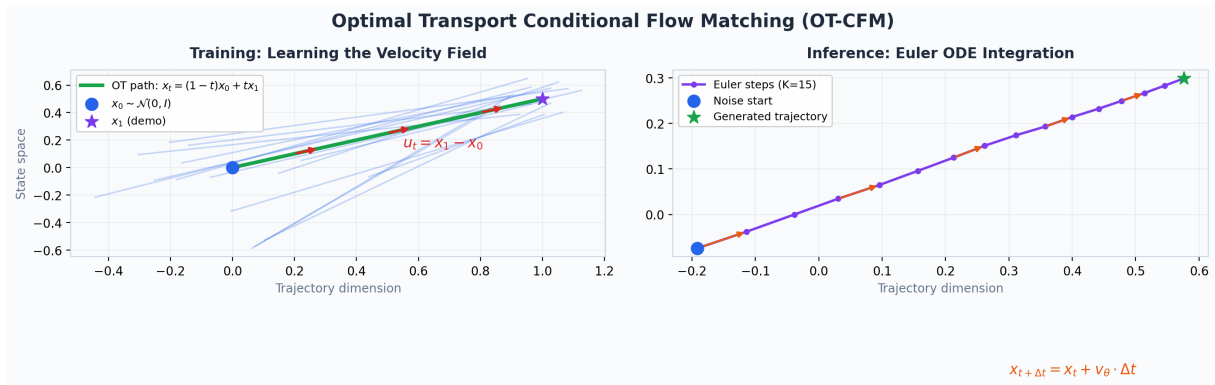


Figure 3: OT-CFM formulation depicting training interpolation and ODE integration at inference.